



# LiteTest

## Test Reference

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This document summarizes the public Test API declared in `include/litetest_test.h`. It focuses on the helper-specific return conventions and the option structs used by the comparison helpers.

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## Preface

This document is intended for LiteTest users and contributors who need quick access to the definitions of terms used in LiteTest or to browse through the terms.

For a list of other LiteTest documents and the LiteTest repository layout, see the LiteTest Documentation Guide.

For a glossary of terms, see the LiteTest Glossary Reference.

## Document Version History

| Document | Runner | Test  | Date       | Comment          | Author/Editor |
|----------|--------|-------|------------|------------------|---------------|
| 1.0      | 1.0.0  | 1.0.0 | 2026-06-11 | Initial version. | Paul Sinclair |

- The **Document** column records the document's version with the format **M.u** (Major, update).
- The **Runner** column records the LiteTest Runner API version current at the time this document version was published and is defined by its `LT_RUNNER_VERSION` macro.
- The **Test** column records the LiteTest Test API version current at the time this document version was published and is defined by its `LT_TEST_VERSION` macro.
- Both Runner and Test use the version format "**M.m.p**" (Major, minor, patch).
- **M** is the same for the Document, Runner, and Test versions.

The document's update version tracks released updates to this document and does not correspond to a minor or patch version. **u** increments when document changes are published in a release without a change to **M**, and it resets to 0 when **M** is incremented.

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## 1. Return Conventions

### 1.1 Return Styles

LiteTest test helpers use three return styles:

- Structured helpers return `LT_OK`, `LT_MISMATCH`, `LT_TIMEOUT`, or an `LT_E*` error code.
- Predicate helpers return 1 for true or match, 0 for false or no match, and -1 on invalid input or runtime error.
- Process helpers that model command completion return 0 for success, 1 for timeout, and -1 for invalid input or setup failure.

### 1.2 Assertion Guidance

Use the helper family to decide what to check:

- File, JSON, metadata, and normalized-text comparisons should usually be checked against `LT_OK` or `LT_MISMATCH`.
- Existence, string, wildcard, and regex predicates should usually be checked against 1 or 0.
- Command-execution helpers should usually be checked against 0 or 1.

## 2. Comparison Option Structs

The comparison helpers use small option structs instead of mixed boolean and flag parameter lists.

### 2.1 `lt_path_compare_options_t`

Used by `lt_compare_paths_with_options`.

```
typedef struct {  
    int flags;  
} lt_path_compare_options_t;
```

Supported flags:

- `LT_FILECMP_IGNORE_TRAILING_WHITESPACE`
- `LT_FILECMP_IGNORE_EMPTY_LINES`
- `LT_FILECMP_IGNORE_TIMESTAMPS`
- `LT_FILECMP_IGNORE_LINE_ENDINGS`
- `LT_FILECMP_CASE_INSENSITIVE`

Default initializer:

```
#define LT_PATH_COMPARE_OPTIONS_INIT {0}
```

### 2.2 `lt_path_metadata_compare_options_t`

Used by `lt_compare_path_metadata`.

```
typedef struct {  
    int flags;  
} lt_path_metadata_compare_options_t;
```

Supported flags:

- `LT_STATCMP_SIZE`
- `LT_STATCMP_PERMS`
- `LT_STATCMP_MTIME`
- `LT_STATCMP_OWNER`
- `LT_STATCMP_TYPE`

Default initializer:

```
#define LT_PATH_METADATA_COMPARE_OPTIONS_INIT {0}
```

Passing `NULL` or the default initializer uses the common default metadata set: size, permissions, and type.

### 2.3 `lt_json_compare_options_t`

Used by `lt_compare_json_with_limit`.

```
typedef struct {  
    size_t max_bytes;
```

```
    int ignore_key_order;
} lt_json_compare_options_t;
```

Default initializer:

```
#define LT_JSON_COMPARE_OPTIONS_INIT {0, 0}
```

Use `ignore_key_order` when semantic JSON equality matters more than source ordering.  
Use `max_bytes` when the caller wants a tighter bound than the implementation default.

## 2.4 `lt_text_compare_options_t`

Used by `lt_compare_text_normalized`.

```
typedef struct {
    int ignore_whitespace;
    int ignore_line_endings;
} lt_text_compare_options_t;
```

Default initializer:

```
#define LT_TEXT_COMPARE_OPTIONS_INIT {0, 0}
```

## 3. Common Usage Patterns

### 3.1 Compare Two Paths While Ignoring Timestamps

```
const lt_path_compare_options_t options = {
    LT_FILECMP_IGNORE_TIMESTAMPS
};
```

```
LT_ASSERT(lt_compare_paths_with_options(expected_path, actual_path, &options) == LT_OK,
```

### 3.2 Compare Path Metadata Including Modification Time

```
const lt_path_metadata_compare_options_t options = {
    LT_STATCMP_SIZE | LT_STATCMP_PERMS | LT_STATCMP_TYPE | LT_STATCMP_MTIME
};
```

```
LT_ASSERT(lt_compare_path_metadata(left_path, right_path, &options) == LT_OK, 0);
```

### 3.3 Compare JSON While Ignoring Object Key Order

```
const lt_json_compare_options_t options = {
    0,
    1
};
```

```
LT_ASSERT(lt_compare_json_with_limit(expected_json, actual_json, &options) == LT_OK, 0);
```

### 3.4 Compare Text While Ignoring Whitespace and Line Endings

```
const lt_text_compare_options_t options = {
    1,
    1
};
```

```
LT_ASSERT(lt_compare_text_normalized(left_text, right_text, &options) == LT_OK, 0);
```

## 4. Convenience Forms

- `lt_compare_paths` performs a byte-for-byte path comparison.
- `lt_compare_json` performs a strict JSON comparison using default limits.
- `lt_compare_file_lines` compares files line-by-line and returns `LT_OK` or `LT_MISMATCH`.

For the full declaration list and per-function comments, see `include/litetest_test.h`.